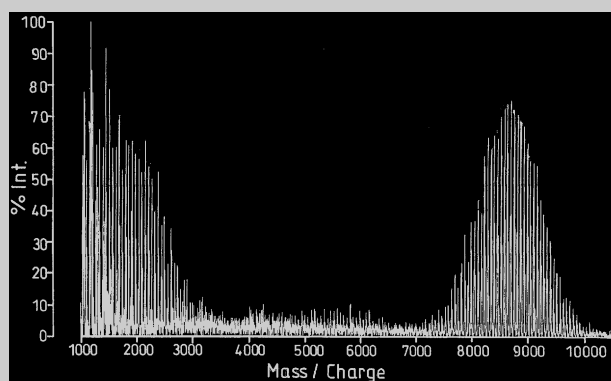


Full Paper: Poly(propylene oxide)s can be analyzed efficiently with respect to functionality and molar mass by coupled liquid chromatography and MALDI-TOF mass spectrometry. Both techniques are coupled via a robotic interface, where the matrix is coaxially added to the eluate and spotted dropwise onto the MALDI target. It is shown that size exclusion chromatography and liquid chromatography at critical conditions of adsorption coupled to MALDI-TOF yield useful structural information on oligomer masses and chemical composition. In particular, the analysis of technical poly(propylene oxide) triols by LC-CC/MALDI-TOF reveals the presence of diols and mono-ols in addition to the triols. The oligomer distributions of all functionality fractions are determined.



Off-line MALDI-TOF spectrum of sample 2.

Matrix-Assisted Laser Desorption/Ionization Mass Spectrometry of Synthetic Polymers, 5^a

Analysis of Poly(propylene oxide)s by Coupled Liquid Chromatography at the Critical Point of Adsorption and MALDI-TOF Mass Spectrometry

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Introduction

Polymers and copolymers of propylene oxide (PPO) are in widespread use in technical applications. The copolymerization of ethylene oxide and propylene oxide yields segmented polymers with amphiphilic properties which are used as surfactants. Glycerol-started PPO is frequently used as a cross-linking polyol for the production of polyurethanes.

In addition to the molar mass distribution (MMD) PPOs are distributed with respect to functional endgroups. This functionality type distribution (FTD) results from the presence of water or other impurities in the reac-

tion mixture. For a proper evaluation of the product properties of PPOs it is, therefore, important to determine FTD as a function of molar mass.

It has been established theoretically and verified experimentally that liquid chromatography at the critical point of adsorption (LC-CC) provides selective information on FTD of functional homopolymers.^[1–5] Different from any other liquid chromatographic technique, LC-CC operates at conditions where the polymer chain does not contribute to retention and the elution behavior of the sample is solely directed by the type and number of the functional endgroups.

Matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry is a particularly effective tool for the molar mass determination of natural and synthetic polymers. The ability to ionize a broad range of materials, its high sensitivity, large mass range,

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fast sample preparation, and the absence of fragmentation are characteristics of this powerful technique. In a MALDI-TOF instrument, short-duration laser pulses are directed at a sample dispersed in a specific matrix. The laser energy causes a portion of the sample/matrix mixture to be desorbed from the surface and ionized. The produced ions are analyzed in a TOF mass analyzer. Initially developed for use with large biomolecules in 1988 by Karas and Hillenkamp,^[6,7] MALDI-TOF has advanced very rapidly into a powerful technique for synthetic polymer analysis.^[8–15]

One major concern is the ability of MALDI-TOF to provide accurate molar mass measurements. It has been shown that for polydispersity indices $\overline{M}_w/\overline{M}_n > 1.1$ there is a significant discrepancy between molar masses calculated from MALDI-TOF vs. that calculated from size exclusion chromatography (SEC). Polymers with higher polydispersities may be completely inaccessible to analysis by MALDI-TOF.^[16–19] A solution to this problem has been proposed recently by combining a SEC prefractionation with a subsequent MALDI-TOF analysis of the resulting fractions.^[16,18,20–23] The feasibility of direct deposition has been shown in ref.,^[22] where the SEC effluent was sprayed onto a moving matrix-coated substrate using a modified LC-Transform Series 100 IR interface of Lab Connections. Alternatively, Nielsen described the coupling of SEC to a robotic interface Probot of Bioanalytical Instruments, where the matrix is coaxially added to the SEC effluent and spotted dropwise onto the MALDI target.^[23]

In this paper, the analysis of functionally heterogeneous PPOs by LC-CC shall be discussed. For the analysis of FTD as a function of molar mass, liquid chromatography shall be interfaced with MALDI-TOF MS. As the liquid chromatographic separation techniques SEC and LC-CC shall be used to demonstrate the differences in the obtainable information from both techniques.

Experimental Part

SEC/LC-CC

The measurements were conducted using a modular system, comprising a Waters HPLC pump Model 510, a Rheodyne six-port injection valve, a Waters refractive index detector Model 410 or an Altech Model 500 ELSD detector. The column for the SEC was Polymer Laboratories PLgel Mixed-E, 300 × 7.8 mm i.d. The eluent was HPLC grade tetrahydrofuran (THF). The column for the LC-CC was Macherey Nagel Nucleosil 5 NH₂, 200 × 4 mm i.d. For these experiments a mobile phase of hexane/isopropyl alcohol was used.

MALDI-TOF-MS

The spectra were recorded on a KRATOS Kompact MALDI 4 instrument. The irradiation source was a pulsed nitrogen laser with a wavelength of 337 nm. The length of one laser

pulse was 3 ns. The measurements were carried out using the following conditions: polarity-positive, flight path-linear, mass-high (20 kV acceleration voltage), 100–150 pulses per spectrum. The delayed extraction technique was used applying delay times of 200–800 ns.

MALDI-TOF Sample Preparation

For interfacing liquid chromatography and MALDI-TOF mass spectrometry a software operated robotic interface Probot of Bioanalytical Instruments (Bensheim, Germany) was used. The eluate stream coming from SEC/LC-CC was deposited dropwise on MALDI-TOF targets. The matrix was added coaxially to the eluate stream. As the matrix dithranol-LiTFA was used.

Samples

Samples 1 and 2 are commercial products of Arcol Co. The mixture of the three poly(ethylene glycol)s was prepared from SEC calibration standards.

Results and Discussion

The experimental setup for the coupled LC/MALDI-TOF experiments is schematically presented in Figure 1. The chromatographic separations are carried out using standard equipment, a refractive index detector is applied for monitoring the separation. The flow rate is 0.5 mL/min. After the chromatography the eluate stream is split with 40 parts of the eluate going to the detector and 1 part going to the Probot interface. With an eluate flow of 12.5 µL/min reaching the interface, the dual collection mode is used to deposit 2 µL/min of the eluate on the MALDI-TOF target. The matrix is coaxially added to the eluate stream before deposition. After the evaporation of the solvent, the MALDI-TOF target is subjected to the spectrometer and spectra are taken from all positions using an automatic scan mode.

In a first experiment, the operation of the experimental setup is checked by measuring a mixture of three standards of poly(ethylene glycol) (PEG). The average molar masses of the PEGs are 4000 g/mol (A), 2000 g/mol (B),

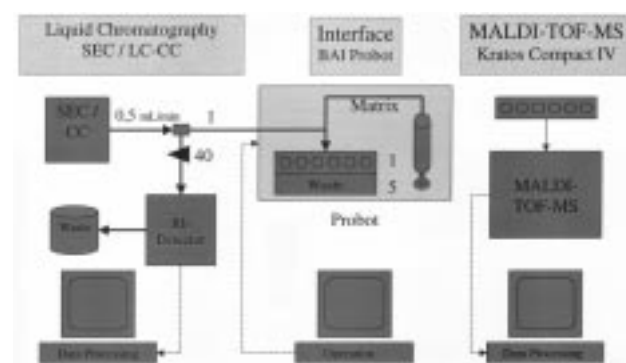


Figure 1. Schematic representation of the coupling of liquid chromatography and MALDI-TOF mass spectrometry.

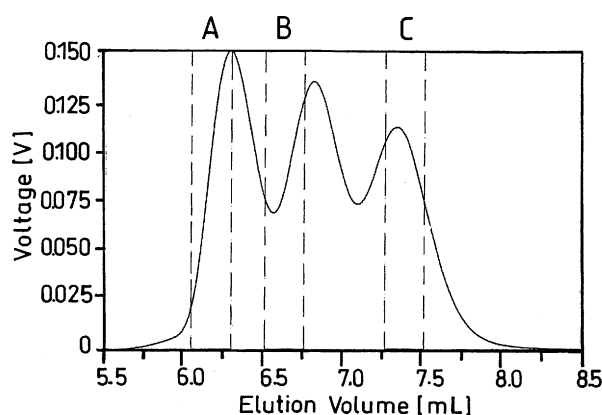


Figure 2. SEC chromatogram of a mixture of three poly(ethylene glycol)s.

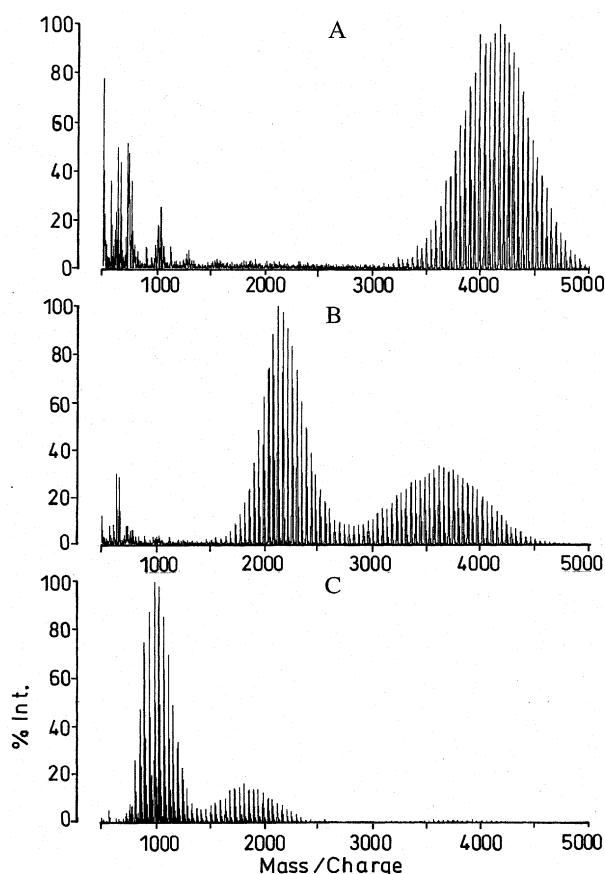
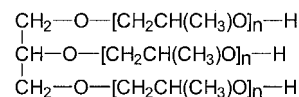


Figure 3. MALDI-TOF spectra of fractions taken from SEC separation of a mixture of three poly(ethylene glycol)s by coupled SEC/MALDI-TOF.

and 1000 g/mol (C), and the separation is carried out in the order of decreasing molar mass by SEC, see Figure 2. The resulting MALDI-TOF spectra for the fractions A–C are presented in Figure 3. As can be seen, well resolved spectra are obtained for all fractions. As the matrix dithranol-LiTFA is used and, therefore, the oligomer mass peaks correspond to the $[M + Li]^+$ molecular ions. Due to the fact, that the fractions are not base line-sepa-

rated in SEC, spectra B and C exhibit residual mass peaks from the previous fractions A and B, respectively.

In the following experiments, poly(propylene oxide)s are analyzed by liquid chromatography and MALDI-TOF mass spectrometry. The idealized chemical structure of glycerol-started PPO is the following:



It is assumed that all three hydroxy groups of the glycerol are equally reactive and, therefore, the length distribution of the PPO polymer chains bound to the primary and secondary hydroxy groups are similar. Due to the presence of small amounts of water in the reaction mixture, very frequently small amounts of poly(propylene glycol) are formed in addition to the PPO-triol.



The MALDI-TOF spectrum of a technical PPO-triol (sample 1) is presented in Figure 4. The average molar mass of the sample is 1500 g/mol which is in very good agreement with the most intense mass peak at 1551 Da, found in the spectrum. The spectrum exhibits one oligomer series which can be assigned to the PPO-triol structure. Since LiTFA has been added to the matrix, cationization takes place via the attachment of Li^+ to form $[M + Li]^+$ molecular ions.

$$[M + Li]^+ = 6.94 (\text{Li}) + 92.10 (\text{Glycerol}) + 58.08 (\text{PO}) \times n$$

n	$[M + Li]^+$ (calc.)	$[M + Li]^+$ (exp.)
23	1434.88	1435
24	1492.96	1493
25	1551.04	1551

The present spectrum does not show any other oligomer series and, therefore, does not indicate other functionality fractions.

In a second step, the sample is separated with respect to molar mass by SEC and different fractions are subjected to MALDI-TOF mass spectrometry via automated fraction collection and deposition on the MALDI-TOF target. The SEC chromatogram and three selected mass spectra are presented in Figure 5. Fraction A contains a high molar mass portion of PPO-triol which has not been detected by the off-line MALDI-TOF experiment. Fraction B is also composed of PPO-triol and exhibits three molar mass fractions centered at 4500, 2800, and 1800 Da. The MALDI-TOF spectrum of the fraction C is very similar to the off-line spectrum in Figure 4 and contains the major amount of the PPO-triols. Finally, the lowest molar mass fraction D shows, in addition to the

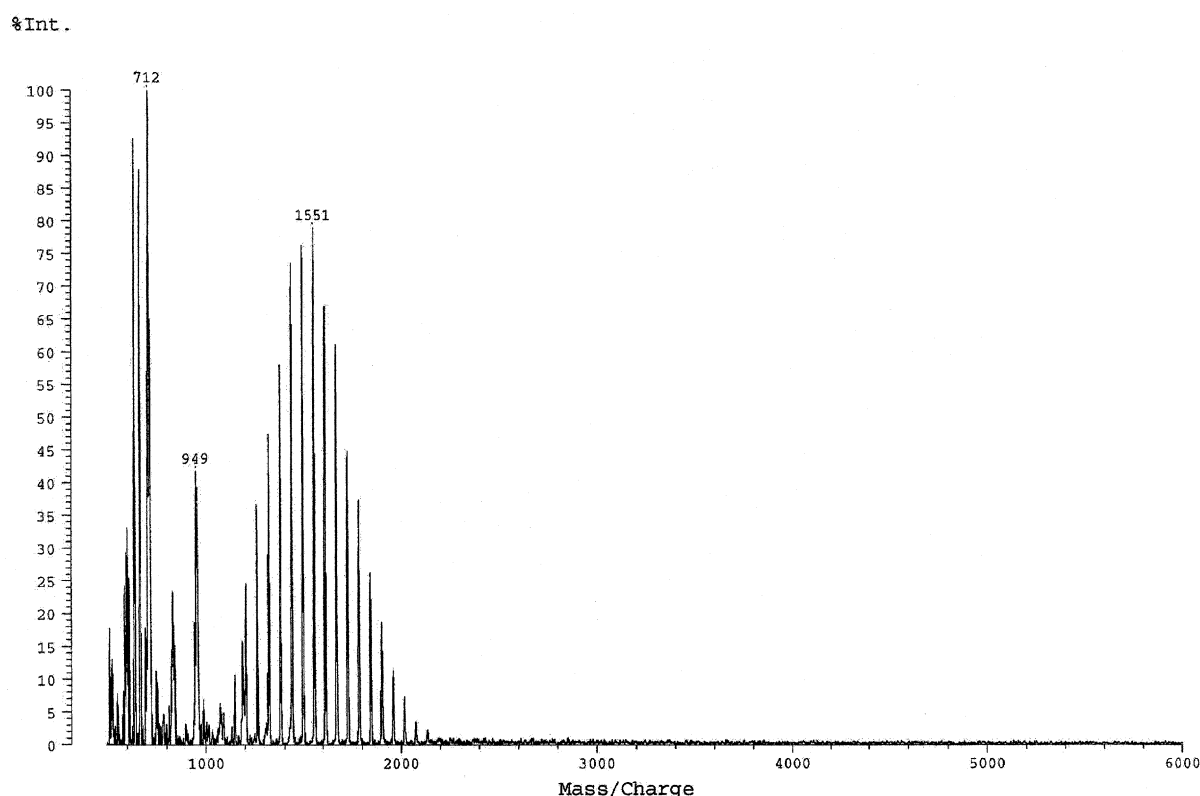


Figure 4. Off-line MALDI-TOF spectrum of sample 1.

oligomer series of the PPO-triols, a second oligomer series centered at 950 Da (indicated by arrow). This oligomer series can be assigned to a small fraction of poly(propylene glycol) (PPO-diol) which has not been detected in the off-line MALDI experiment. Due to the overlapping with the PPO-triol oligomer series, the PPG peaks can barely be recognized. The peak masses of this series correspond to $[M + Li]^+ = 25 + 58n$, where n is the degree of polymerization.

Much more selectively than SEC, LC-CC can be used to obtain information on the structural peculiarities of the sample. As was already mentioned, LC-CC separates functionally heterogeneous samples with regard to their functional groups irrespective of the molar mass. For this type of experiments, in a first step the critical conditions for the respective homopolymer, i.e., PPO, must be determined. This can be done by running PPGs with different molar masses in binary eluents of varying composition. In the present case, the critical point of adsorption for PPG is obtained at an eluent composition of hexane/isopropyl alcohol 92:8 vol.-%. The functionality type separation of sample 1 under these conditions is shown in Figure 6. The LC-CC chromatogram exhibits three peaks at elution volumes of 3.3 mL, 5.1 mL, and 8.2 mL (the peak at 2.4 mL is the injection peak).

Similar to the SEC experiment, the LC-CC elution peaks are analyzed by direct coupling to MALDI-TOF MS. While for the first fraction a spectrum cannot be

obtained due to very low concentration, well resolved spectra for peaks B and C are shown in Figure 6. Different from SEC, the LC-CC peaks are uniform with respect to chemical composition (functionality). Peak B can be identified as the diol (PPG) fraction, while peak C is due to the PPO-triol oligomers. In agreement with the SEC/MALDI-TOF results the PPG fraction is centered at 950 Da, see spectrum B. The PPO-triol fraction exhibits a bimodal oligomer distribution, the major distribution being centered at 1500 Da. The second distribution is centered at 4500 Da and is very low in concentration. To summarize, using the combination of LC-CC separation and MALDI-TOF detection it is possible to identify PPG as a by-product in the PPO-triol and to obtain information on the oligomer distribution of both functionality fractions. While the PPG exhibits a monomodal oligomer distribution, the PPO-triol is composed of two fractions with different molar masses.

The off-line MALDI-TOF spectrum of a second technical PPO-triol (sample 2) with an average molar mass of 8500 g/mol is presented in Figure 7. Three different molar mass fractions centered around 9000, 6000 and 2000 Da are obtained, the highest molar mass fraction being the PPO-triol. For a more detailed insight, the sample is fractionated by SEC and fractions A-D analyzed by MALDI-TOF mass spectrometry. Fraction A, C, and D are presented in Figure 8. As is shown in Figure 8 A, fraction A exhibits two oligomer distributions. The higher

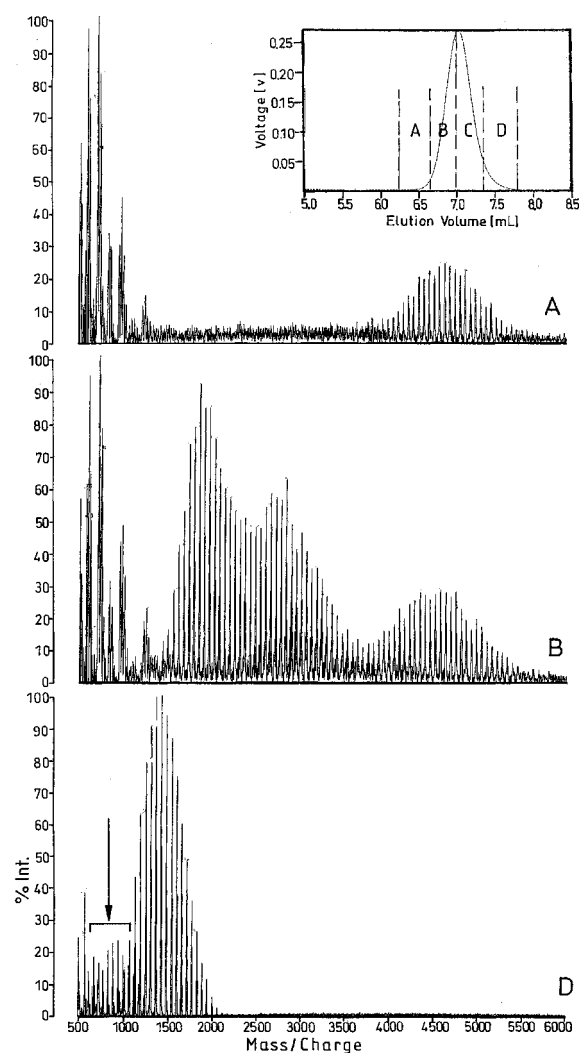
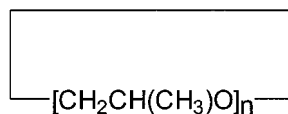
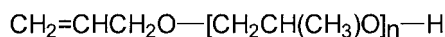


Figure 5. SEC chromatogram of sample 1 and MALDI-TOF spectra of fractions obtained by coupled SEC/MALDI-TOF.

molar mass peak series can be identified as PPO-triol, while the peak series centered around 5500 Da belongs to PPG. In fraction C, a low molar mass fraction of PPO-triol is observed, and in addition an oligomer series appears (indicated by arrow) which cannot be attributed to PPO-triol or PPG. The observed peak series ... 2214–2272–2330 ... can be assigned to PPOs with allylic end-groups or cyclic PPO oligomers.



Unfortunately, MALDI-TOF mass spectrometry is not able to differentiate the two different structures because their calculated oligomer masses do not deviate from each other.

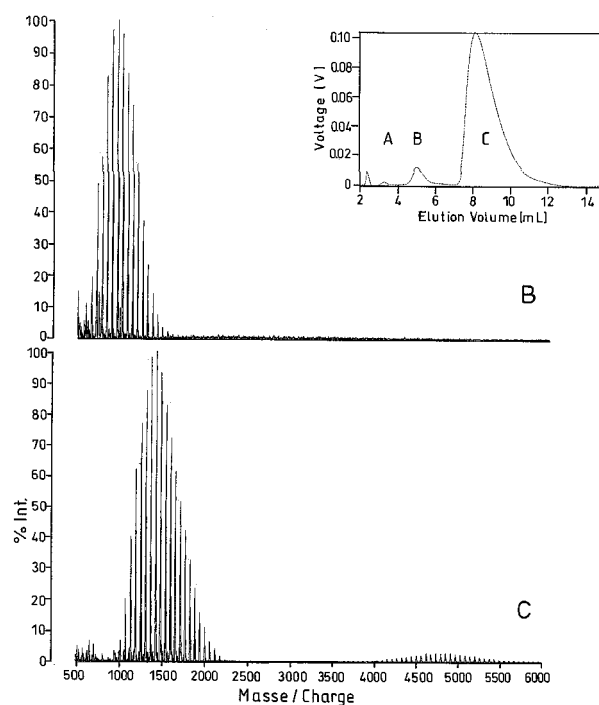


Figure 6. LC-CC chromatogram of sample 1 and MALDI-TOF spectra of fractions obtained by coupled LC-CC/MALDI-TOF.

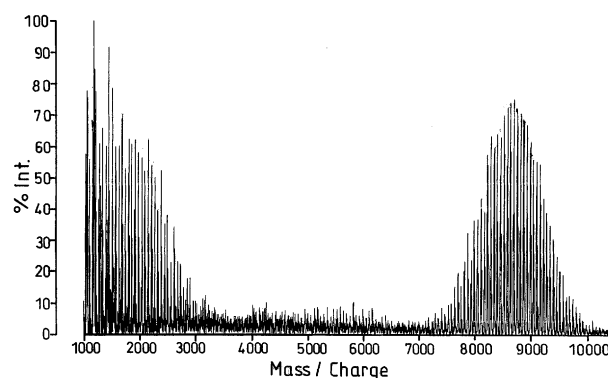


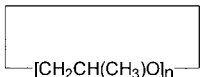
Figure 7. Off-line MALDI-TOF spectrum of sample 2.

Allylics		Cyclics	
$[\text{M}+\text{Li}]^+ = 6.94 + 58.08 + 58.08 n$		$[\text{M} + \text{Li}]^+ = 6.94 + 58.08 n$	
n	$[\text{M}+\text{Li}]^+ \text{ (calc.)}$		$[\text{M}+\text{Li}]^+ \text{ (calc.)}$
37	2213.98		2155.90
38	2272.06		2213.98
39	2330.14		2272.06

Fraction D mainly contains PPOs with allylic end-groups and/or cyclic oligomers.

As has been shown previously, the prefractionation of the sample by LC-CC is the most versatile approach for a complete analysis. The LC-CC chromatogram of sample 2 in Figure 9 indicates that the sample contains three different functionality fractions. In agreement with the results for sample 1, the latest eluting fraction can be assigned to the PPO-triol. The fraction eluting at an elu-

Table 1. Analysis of sample 2 by coupled LC-CC and MALDI-TOF mass spectrometry.

LC-CC fraction	A	B	C + D
Structure	$\text{CH}_2=\text{CHCH}_2\text{O}-[\text{CH}_2\text{CH}(\text{CH}_3)\text{O}]_n-\text{H}$  $[\text{M} + \text{Li}]^+ = 6.94 + 58.08 n$	$\text{HO}-[\text{CH}_2\text{CH}(\text{CH}_3)\text{O}]_n-\text{H}$ $[\text{M} + \text{Li}]^+ = 24.96 + 58.08 n$	$\text{CH}_2=\text{O}-[\text{CH}_2\text{CH}(\text{CH}_3)\text{O}]_n-\text{H}$ $\text{CH}=\text{O}-[\text{CH}_2\text{CH}(\text{CH}_3)\text{O}]_n-\text{H}$ $\text{CH}_2=\text{O}-[\text{CH}_2\text{CH}(\text{CH}_3)\text{O}]_n-\text{H}$ $[\text{M} + \text{Li}]^+ = 99.04 + 58.08 n$
Molar mass			
Oligomer series ($\bar{M}_p$)	400–3 200 Da (800)	800–3 000 (1 800) 4 900–6 400 (5 600)	1 300–2 000 (1 500) 2 400–3 600 (3 100) 7 000–10 000 (8 600)

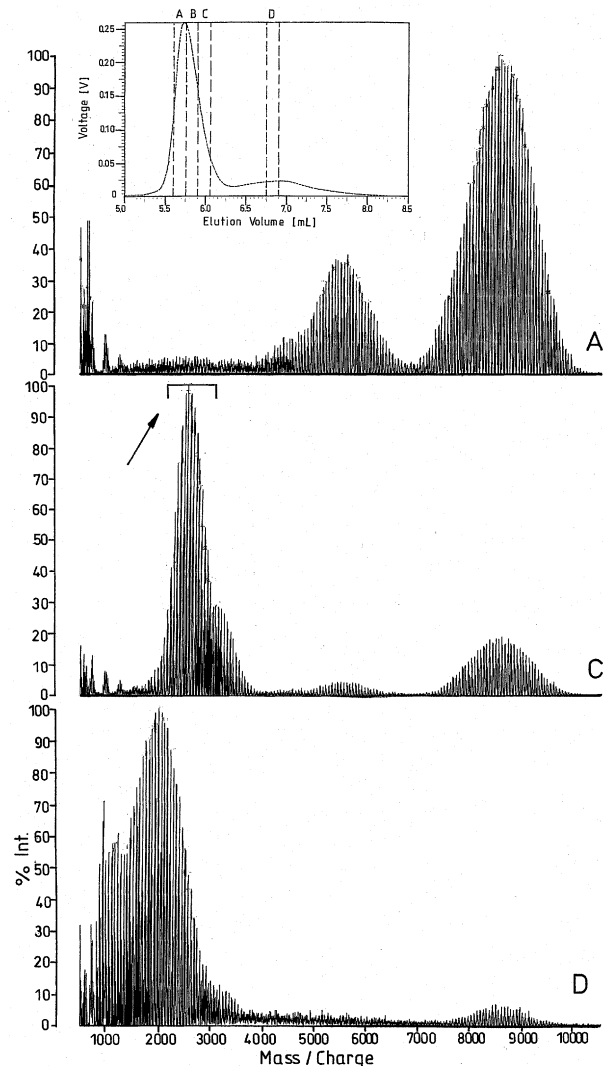


Figure 8. SEC chromatogram of sample 2 and MALDI-TOF spectra of fractions obtained by coupled SEC/MALDI-TOF.

tion volume of 5 mL corresponds to PPG. The fraction eluting at the lowest elution volume is significantly less polar than PPO-triol and PPG. This fraction is tentatively assigned to allyl-terminated PPO and cyclic oligomers.

The analysis of the LC-CC fractions by MALDI-TOF reveals all molar mass peculiarities of the fractions. The allylic/cyclic fraction in Figure 9A exhibits one oligomer

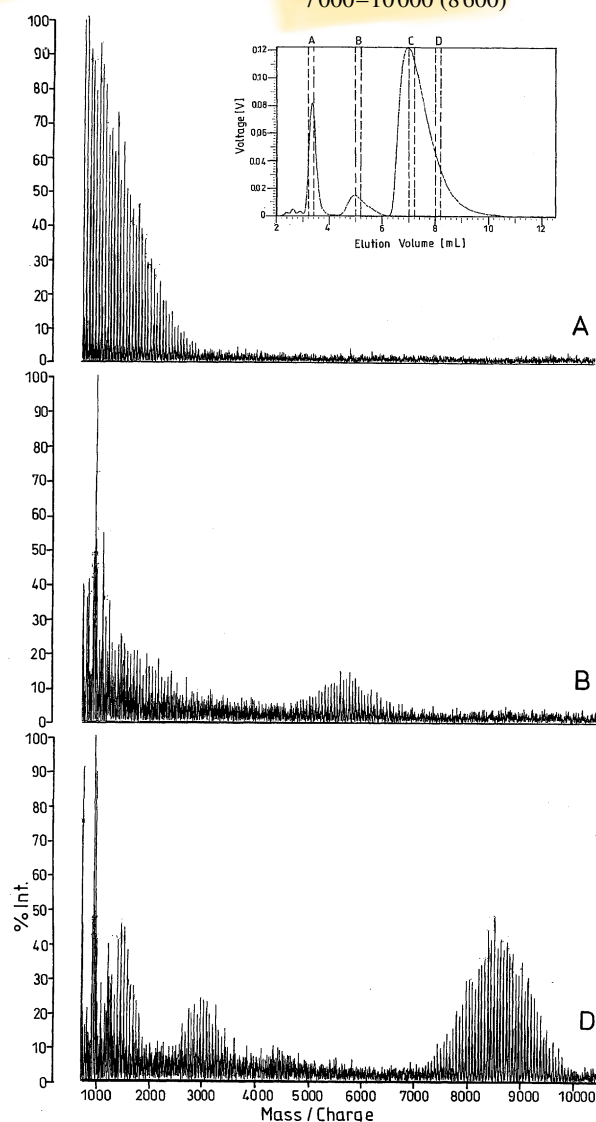


Figure 9. LC-CC chromatogram of sample 2 and MALDI-TOF spectra of fractions obtained by coupled LC-CC/MALDI-TOF.

distribution centered at around 800 Da. The PPG fraction in Figure 9B is bimodally distributed, one distribution centered at 5600 Da while the other distribution is centered at 1800 Da. Finally, the PPO-triol fraction in Figure

9D exhibits three oligomer series centered at 8600 Da, 3100 Da, and 1500 Da. The total structural information obtained for sample 2 by coupled LC-CC and MALDI-TOF mass spectrometry is summarized in Table 1.

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